A Convention-Closed Darboux–Halphen Atlas: Theta Series, Modular Covariance, and Collision Strata

Route-A source-local certificate HCS-C320

3 September 2026

Abstract

We lock a polynomial, nome, and theta convention for the Darboux–Halphen system and prove that the logarithmic derivatives of the three Jacobi theta constants solve it. Their exact series also give an Eisenstein divisor-sum bridge.

1 Polynomial convention and theta/q-series lock

Let a prime mean $\mathrm{d}/\mathrm{d}\tau$ and fix

$$x_1' = x_2x_3 - x_1(x_2+x_3), \quad x_2' = x_3x_1 - x_2(x_3+x_1), \quad x_3' = x_1x_2 - x_3(x_1+x_2). \tag{1}$$

For $\tau \in \mathbb{H}$, set $Q = e^{\pi \mathrm{i}\tau}$ and use theta constants with this nome. Define

$$(x_1, x_2, x_3) = -2\frac{\mathrm{d}}{\mathrm{d}\tau}(\log\vartheta_2, \log\vartheta_3, \log\vartheta_4). \tag{2}$$

Theorem 1 (Theta seed, exact series, and arithmetic bridge)**.** *The triple (2) solves (1). Put $D = Q\,\mathrm{d}/\mathrm{d}Q$ and $X_j = x_j/(\pi\mathrm{i})$. Then $DX_1 = X_2X_3 - X_1(X_2+X_3)$, cyclically, and*

$$\begin{aligned}\vartheta_2 &= 2Q^{1/4}\sum_{m\ge0} Q^{m(m+1)},\\ \vartheta_3 &= 1 + 2\sum_{m\ge1} Q^{m^2}, \qquad \vartheta_4 = 1 + 2\sum_{m\ge1}(-1)^m Q^{m^2}.\end{aligned} \tag{3}$$

In particular $X(0) = (-1/2, 0, 0)$. Moreover

$$X_1 + X_2 + X_3 = -\frac{1}{2}E_2(\tau) = -\frac{1}{2} + 12\sum_{n\ge1}\sigma_1(n)Q^{2n}. \tag{4}$$

Proof. Jacobi's triple products give (3). Their logarithmic derivatives converge normally on compact subsets of $|Q| < 1$. The standard theta derivative identities are

$$\begin{aligned}\frac{1}{\pi\mathrm{i}}(\log\vartheta_2)' &= \frac{1}{12}(E_2 + \vartheta_3^4 + \vartheta_4^4),\\ \frac{1}{\pi\mathrm{i}}(\log\vartheta_3)' &= \frac{1}{12}(E_2 + \vartheta_2^4 - \vartheta_4^4),\\ \frac{1}{\pi\mathrm{i}}(\log\vartheta_4)' &= \frac{1}{12}(E_2 - \vartheta_2^4 - \vartheta_3^4).\end{aligned}$$

Substitution, using $\vartheta_3^4 = \vartheta_2^4 + \vartheta_4^4$,

$$E_4 = \tfrac{1}{2}(\vartheta_2^8 + \vartheta_3^8 + \vartheta_4^8), \qquad DE_2 = \tfrac{1}{6}(E_2^2 - E_4)$$

for the present square-root nome Q, reduces each residual in (1) to zero. Since $\mathrm{d}/\mathrm{d}\tau = \pi\mathrm{i}D$, the scaled polynomial law follows without an omitted factor. Finally $\vartheta_2\vartheta_3\vartheta_4 = 2\eta^3$ and $(\log\eta)' = \pi\mathrm{i}E_2/12$ give (4); its last equality is the standard E_2 series in Q^2. □

Equation (4) is an intrinsic quasimodular divisor-sum identity. It is neither a target Λ-function nor a prime-orbit correspondence.

Source lineage

References

[1] Y. Ohyama, "Differential Relations of Theta Functions," *Osaka Journal of Mathematics* 32 (1995), 431–450. Repository copy: ocu-omu.repo.nii.ac.jp.

[2] J. A. Cruz Morales, H. Movasati, Y. Nikdelan, R. Roychowdhury, and M. A. C. Torres, "Manifold Ways to Darboux–Halphen System," *SIGMA* 14 (2018), 003. DOI: 10.3842/SIGMA.2018.003.

[3] J. Harnad and J. McKay, "Modular Invariants and Generalized Halphen Systems," arXiv: solv-int/9902012.